Technical Guide: Implementing the AspeQt-2k26 Modem Bridge

Subtitle: Connecting Atari 850 Hardware to the Modern Internet via RS-232

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Project Repo: https://github.com/pjones1063/AspeQt-2k26

1. Overview

[IMPORTANT NOTE REGARDING THE R: DEVICE]

This document details how to use the Legacy RS-232 Modem Bridge, which requires connecting physical Atari hardware (like an 850 Interface or P:R: Connection) to your PC.

Note: AspeQt-2k26 also features a pure software R: Device (850 Emulation) that requires no vintage hardware at all. We previously believed this software mode was only possible using the hardware-level interrupts of a Raspberry Pi GPIO connection. However, it is fully compatible with standard PC USB FTDI adapters, provided the CTS logic is properly inverted.

If you are looking to set up the software-only R: Device without physical Atari RS-232 hardware, please refer to the Raspberry Pi Technical Guide or the SIO2PC USB Interface Guide instead.

https://github.com/pjones1063/AspeQt-2k26/blob/main/doc/SIO2PC_Build_Instructions.pdf

The AspeQt-2k26 Modem Bridge is a hardware-software hybrid solution that allows vintage Atari 8-bit computers to access modern Telnet BBSs and TCP/IP services. Unlike pure software emulators (like the R: Device), this feature utilizes the **physical Atari 850 Interface** (or compatible RS-232 hardware like the P:R: Connection or ICD R-Verter).

AspeQt acts as a "Virtual Hayes Modem," bridging the physical serial data from the Atari to a TCP/IP socket on the PC. This allows the use of authentic vintage terminal software (BobTerm, Ice-T, Express!) to connect to the internet.

2. Hardware Architecture

To utilize the Modem Bridge, your setup requires **two distinct connections** to the PC. One handles standard disk I/O (SIO), and the other handles the Modem traffic (RS-232).

The "Dual-Cable" Topology

1. **Disk I/O Path:** Atari SIO -> SIO2PC -> PC USB (COM A)
    - *Function:* Handles D1: through D8:
2. **Modem Path:** Atari 850 (Port 1) -> Null Modem -> PC USB (COM B)
    - *Function:* Handles R: (The Modem Bridge)

Logical Diagram:

```
       [ ATARI COMPUTER ]                |                 [ PC RUNNING ASPEQT ]
------------------------------------------------------------------------------------------

      +-------------------+                                  +----------------------+
      |                   |            (1)                   |                      |
      |     SIO PORT      |========[ SIO2PC ]==============>|   [ USB PORT 1 ]     |
      |                   |   ( Daisy Chain SIO Cable )      |   (SIO / Disks)      |
      +-------------------+                                  +----------------------+
              |                                                        |
              |                                                        v
              | ( Daisy Chain SIO Cable )                    ( ATR, ATX, XEX, TNFS )
              |
              |
      +-------------------+                                  +----------------------+
      |   850 INTERFACE   |            (2)                   |                      |
      |    (R: DEVICE)    |======[ CUSTOM NULL ]===========>|   [ USB PORT 2 ]     |
      |      PORT 1       |        MODEM CABLE               |   (Modem Bridge)     |
      +-------------------+                                  +----------------------+
                                                                       |
                                                                       v
                                                               ( THE INTERNET )
```

3. The Critical Link: Null Modem Wiring

Standard off-the-shelf Null Modem cables **will not work** with the Atari 850 without modification. The 850 requires the **DSR** (Data Set Ready) and **CD** (Carrier Detect) pins to be held HIGH to detect the presence of a modem.

AspeQt asserts the PC's **DTR** (Data Terminal Ready) signal when the bridge is active. You must construct a cable that routes this DTR signal to the 850's DSR/CD pins.

Wiring Pinout (DB9 to DB9)

Atari 850 (Male)	**Direction**	**Connection Type**	**PC USB Adapter (Female)**
Pin 1 (DTR)	Out ->	**Cross**	**Pin 6 (DSR)** + **Pin 1 (CD)**
Pin 6 (DSR) + **Pin 2 (CD)**	<- In	**Cross**	**Pin 4 (DTR)**
Pin 3 (TX)	Out ->	**Cross**	**Pin 2 (RX)**
Pin 4 (RX)	<- In	**Cross**	**Pin 3 (TX)**
Pin 7 (RTS)	Out ->	**Cross**	**Pin 8 (CTS)**
Pin 8 (CTS)	<- In	**Cross**	**Pin 7 (RTS)**
Pin 5 (GND)	--	**Direct**	**Pin 5 (GND)**

Note on Loopbacks: *

On the PC side: Bridge Pin 1 (CD) and Pin 6 (DSR) together, and connect them to the wire coming from the Atari's Pin 1 (DTR). This ensures the PC knows the Atari is ready.
On the Atari side: Bridge Pin 2 (CD) and Pin 6 (DSR) together, and connect them to the wire coming from the PC's Pin 4 (DTR). This ensures the Atari 850 detects a valid carrier tone and opens the port for data.

4. AspeQt Configuration

Once the hardware is connected, configure the software bridge:

1. **Open Settings:** Go to Tools -> Options -> Modem Bridge.
2. **Select Port:** Choose the COM port corresponding to your **RS-232 Adapter** (NOT the SIO2PC adapter).

3. **Baud Rate:** Set to **19200**.
    - *Tip:* While the bridge supports higher speeds, 19200 is the sweet spot for stability on standard 850 interfaces.
4. **Flow Control:** Uncheck (OFF).
    - *Reason:* AspeQt handles flow internally. Hardware flow control can cause hangs if the cable wiring is imperfect.
5. **Enable:** Check "Enable Modem Bridge" and click **OK**.

Verification: If wired correctly, the **DTR LED** on your USB adapter should light up immediately upon clicking OK.

Configuration Reference

Configure the bridge settings via **Tools** > **Options** > **Modem Bridge**. Correct settings are vital for stable communication.

Setting	Description & Recommended Value
Enable Modem Bridge	Master switch for the feature. When checked, AspeQt opens the serial port and asserts DTR/RTS signals. *Note: If unchecked, the Atari will see the modem as "Offline" or "Not Ready".*
Serial Port	Select the COM port (Windows) or TTY device (Linux/Mac) corresponding to your **USB-to-Serial adapter**. Do NOT select your SIO2PC adapter here.
Baud Rate	The communication speed between PC and Atari. Must match your terminal software settings. **Recommended:** `19200` (Standard for Atari 850).
Flow Control	Enables hardware RTS/CTS handshaking. **Recommended: OFF** (Unchecked) unless you have a high-quality 5-wire Null Modem cable. If enabled with a 3-wire cable, the connection will freeze.
Local Echo	If checked, AspeQt echoes all typed characters back to the Atari. Use this if you cannot see what you are typing in your terminal software. **Recommended: OFF** (Most BBS software handles echoing remotely).
Phonebook File	Path to the `phonebook.xml` file storing your BBS list. Defaults to the application directory.

5. Using the Phonebook

The Phonebook is essential for managing BBS addresses and storing credentials for the macro system.

1. **Access:** Tools -> Phonebook.
2. **Add Entry:** Click Add.
    - **Name:** A recognizable name (e.g., "Basement BBS").
    - **Address:** The URL or IP (e.g., basementbbs.ddns.net).
    - **Port:** Usually 23.
    - **Login/Password:** (Optional) Enter your BBS credentials here to use the Auto-Type macros.
3. **Dialing:** Select an entry and click Dial.
    - AspeQt will send the ATDT string to the Atari automatically.

Getting a Phonebook: You can download a compatible XML or SyncTerm phonebook list from the Telnet BBS Guide: https://telnetbbsguide.com/lists/new-bbs-systems/

6. Hayes Emulation & Command Set

The bridge emulates a standard Hayes-compatible modem. You can issue these commands from any Atari terminal program.

Standard Commands

Command	Description
AT	**Attention.** Returns OK. Used to verify connection.
ATDT <Name>	**Dial by Name.** Example: ATDT Basement. Looks up IP/Port in Phonebook and loads credentials.
ATDT <Address>	**Raw Dial.** Example: ATDT 192.168.1.5:23. Connects directly (No macros available).
ATH	**Hangup.** Disconnects the TCP session.
+++	**Escape.** Pauses data mode and switches to command mode. Requires 1 second silence before/after.

Special Macros & Features

To overcome the limitations of the Atari keyboard, the bridge provides the following hotkeys during a session:

Key Combo	Function
ESC + U	**Auto-User.** Types the Username saved in the current Phonebook entry.
ESC + P	**Auto-Pass.** Types the Password saved in the current Phonebook entry.
ESC + H	**Panic Hangup.** Immediately terminates the connection and forces Command Mode.

ATASCII Translation

Modern servers expect ASCII. The bridge automatically performs these translations:

- **Delete Key:** Atari Delete (126/127) is converted to ASCII Backspace (0x08).
- **EOL:** Atari EOL (155) is converted to CR/LF.

7. Troubleshooting

"Modem Not Responding" in Terminal

- **Cause:** The Atari 850 does not detect the DSR/CD signal.
- **Fix:** Verify the wiring in Section 3. Ensure AspeQt "Enable Modem Bridge" is checked.

"Garbage Characters" on Screen

- **Cause:** Baud rate mismatch.
- **Fix:** Ensure the baud rate in AspeQt Options matches the baud rate configured in your Atari terminal software (e.g., 19200).

"Session Frozen / Hanging"

- **Cause:** Telnet flow control deadlock.
- **Fix:** Press **ESC** + **H** to force a disconnect, then type ATZ to reset the bridge.

8. Raspberry Pi Implementation (Direct UART)

This section details the configuration for users running AspeQt-2k26 directly on a Raspberry Pi. In this configuration, the Raspberry Pi acts as both the host for the emulator and the physical modem interface.

8.1 Hardware Architecture (Pi Edition)

Unlike the PC setup which requires two USB cables, this setup utilizes the Pi's built-in GPIO pins for the modem connection.

- **Disk I/O Path:** Atari SIO → SIO2PC → **Raspberry Pi USB Port**
  - *Function:* Handles standard disk drives (D1:-D8:).
- **Modem Path:** Atari 850 (Port 1) → Level Shifter → **Raspberry Pi GPIO Header**
  - *Function:* Handles R: (The Modem Bridge).

8.2 The "Voltage Hazard" Warning

* * **CRITICAL:** The Atari 850 uses RS-232 voltage levels (±12V). The Raspberry Pi GPIO uses 3.3V TTL logic.

- **DO NOT** connect the Atari 850 directly to the Raspberry Pi.
- **YOU MUST** use a 3.3V RS-232 to TTL Level Shifter (e.g., MAX3232) to bridge the connection. Direct connection will permanently damage the Pi.

8.3 Wiring: Pi GPIO to Atari 850 (Port 1)

Since the Raspberry Pi UART lacks standard DTR/DSR control pins, we utilize a "Local Loopback" on the Atari side to trick the 850 into detecting a "Ready" modem.

Pinout Table (MAX3232 Shifter Intermediate)

Signal Function	Atari 850 (DB9 Pin)	Direction	Level Shifter (TTL Side)	Raspberry Pi GPIO
Transmit Data	**Pin 3** (TX)	Atari → Pi	T2IN → T2OUT	**GPIO 15** (RXD) - Pin 10
Receive Data	**Pin 4** (RX)	Pi → Atari	R2OUT ← R2IN	**GPIO 14** (TXD) - Pin 8
Ground	**Pin 5** (GND)	Common	GND	**GND** - Pin 6
Clear To Send	**Pin 8** (CTS)	Pi → Atari	T1IN → T1OUT	**GPIO 17** (RTS) - Pin 11
Request To Send	**Pin 7** (RTS)	Atari → Pi	R1OUT ← R1IN	**GPIO 16** (CTS) - Pin 36
Modem Ready	**Pin 6** (DSR)	**Loopback**	--	--
Carrier Detect	**Pin 2** (CD)	**Loopback**	--	--
Terminal Ready	**Pin 1** (DTR)	**Loopback**	--	--

The "Always Ready" Loopback: Inside the DB9 connector plugging into the Atari 850, solder a small jumper wire connecting **Pin 1 (DTR)** to both **Pin 6 (DSR)** and **Pin 2 (CD)**. This ensures the Atari 850 instantly sees the "modem" (Pi) as online when the Atari is powered on.

8.4 Raspberry Pi OS Configuration

Before running AspeQt, you must free the UART from the Linux console.

1. **Disable Serial Console:**
   - Run `sudo raspi-config`
   - Navigate to **Interface Options** → **Serial Port**.
   - "Would you like a login shell to be accessible over serial?" → **No**.
   - "Would you like the serial port hardware to be enabled?" → **Yes**.
2. **Optimize UART (Disable Bluetooth):**
   - To ensure stable timing for the Atari 850, use the primary UART (`PL011`) by disabling Bluetooth.
   - Edit config: `sudo nano /boot/config.txt`

- Add line: `dtoverlay=disable-bt`
- Save and Reboot.

8.5 AspeQt Configuration (Pi Specifics)

Change the settings in **Tools > Options > Modem Bridge** to match the Linux environment:

Setting	Value for Raspberry Pi
Serial Port	`/dev/ttyAMA0` (if BT disabled) OR `/dev/serial0`
Baud Rate	**19200** (Recommended)
Flow Control	**OFF** (Recommended for initial setup). Only enable if the "Loopback" wiring above is 100% verified.